Task 3: Sales Forecasting Model

Alfido Tech Business Analytics Internship **|** Forecasting electricity demand as the required time-series business-demand use case

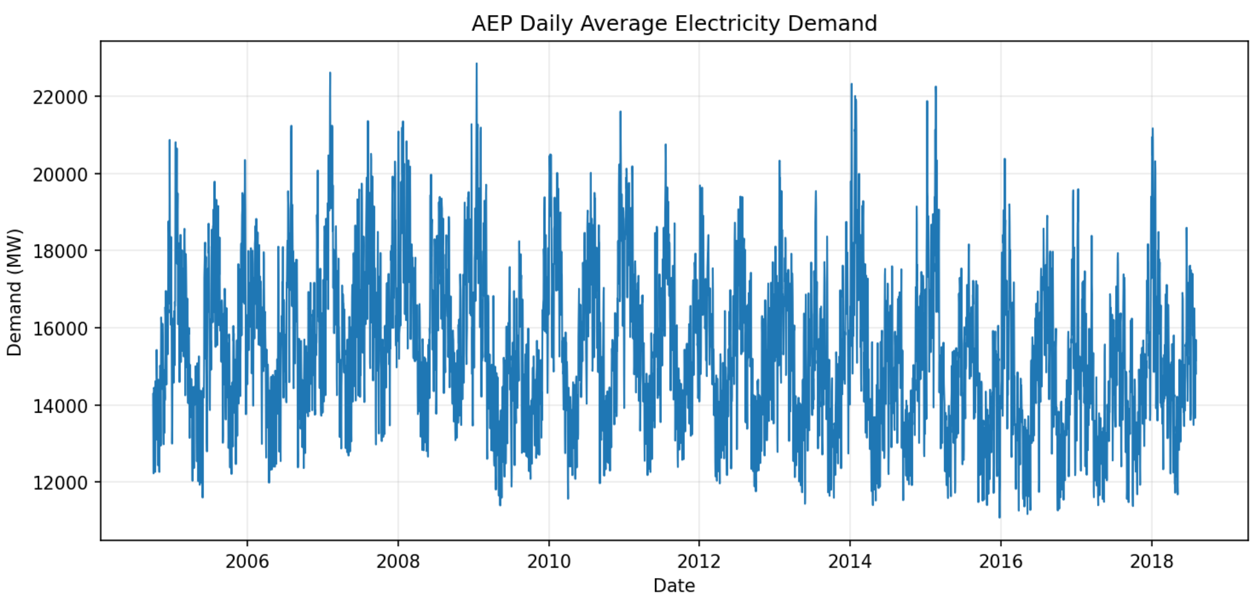
Executive Summary

The hourly AEP demand series was converted into a clean daily time series and modeled with a seasonal SARIMAX model. A 90-day holdout evaluation produced an MAE of **882 MW**, MAPE of **5.48%**, and RMSE of **1,237 MW**. The final model was then refit on the complete history and used to produce a 30-day baseline forecast with 95% confidence intervals and ±5% planning scenarios.

1. Data Preparation

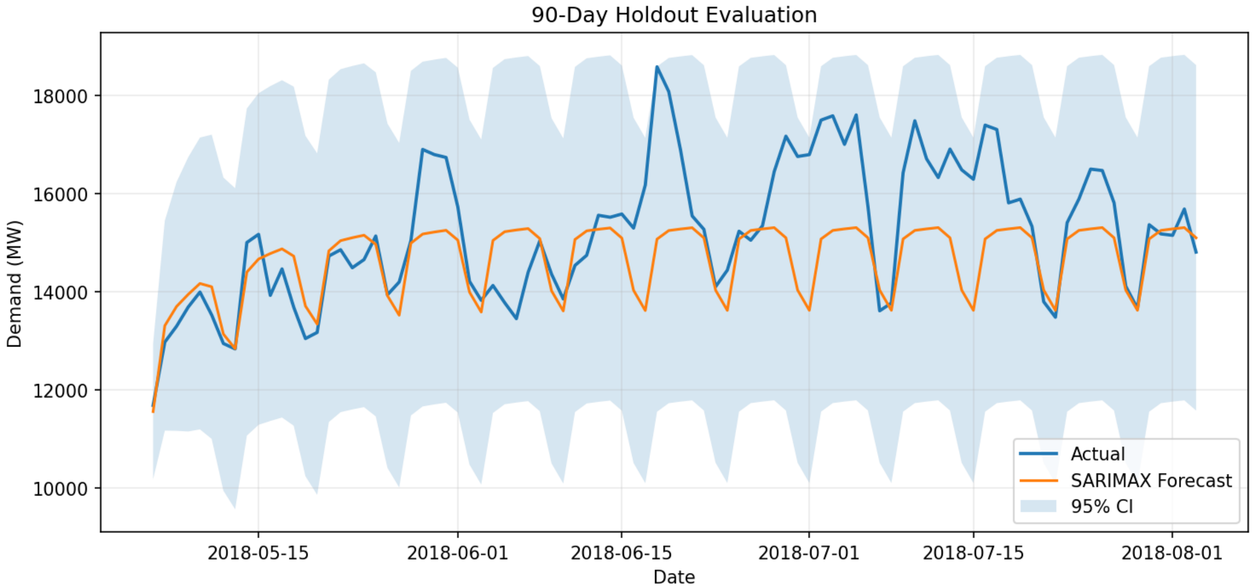
Metric	Result
Time range	01 Oct 2004 to 03 Aug 2018
Original hourly rows	121,273
Duplicate timestamps	4
Missing hourly timestamps before interpolation	27
Model frequency	Daily average demand
Daily observations	5,055

Cleaning steps: parse timestamps, coerce demand to numeric, average duplicate timestamps, rebuild a complete hourly index, interpolate missing hourly observations, then resample to daily mean demand. This preserves the time-series structure while making the data suitable for forecasting.



2. Forecasting Method

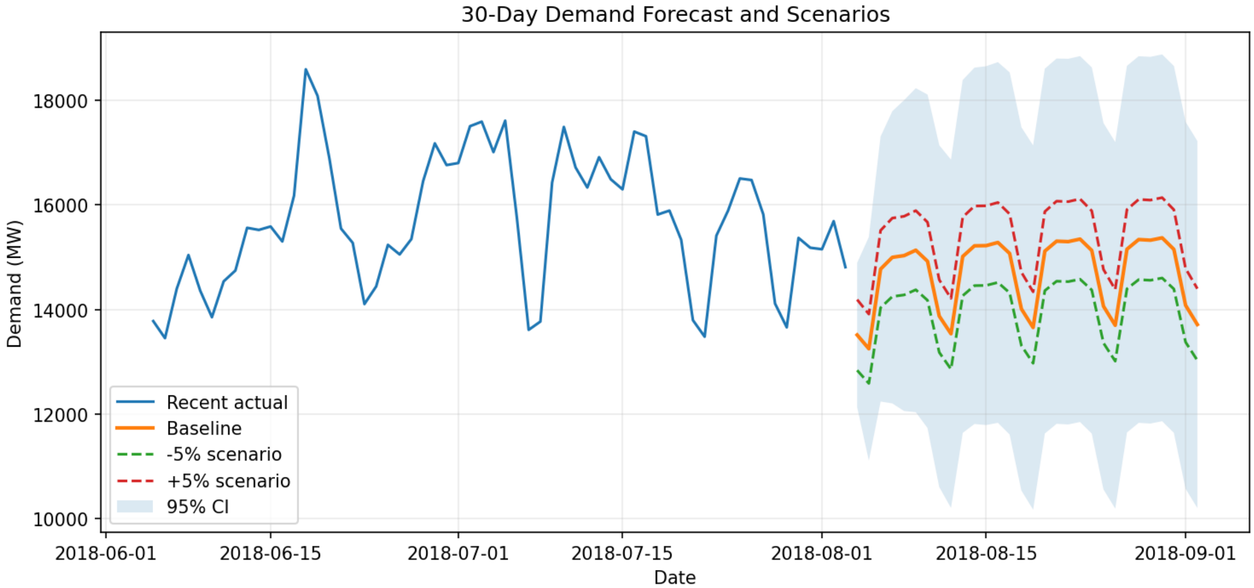
The selected model is SARIMAX(2,0,1) × (1,1,1,7). The daily series uses a seasonal period of 7 to represent weekly demand patterns. The last 90 days were withheld for out-of-sample evaluation. After evaluation, the model was refit on all cleaned observations for the final 30-day forecast.



Evaluation metric	Value	Interpretation
MAE	882 MW	Average absolute forecast error
MAPE	5.48%	Average percentage error
RMSE	1,237 MW	Penalizes larger errors more strongly

3. 30-Day Scenario Forecast

The baseline is the model's point forecast. The low and high planning scenarios apply -5% and +5% adjustments to the baseline. The 95% confidence interval represents statistical forecast uncertainty and should not be interpreted as a guaranteed business range.



Forecast statistic	Value
Average 30-day baseline	14,685 MW
Minimum baseline	13,249 MW
Maximum baseline	15,369 MW
Average low scenario (-5%)	13,951 MW
Average high scenario (+5%)	15,420 MW

4. Business Actions

1. Use the baseline forecast as the near-term operating plan for demand/capacity decisions.

2. Prepare contingency capacity or staffing against the upper confidence bound and the +5% scenario.

3. Use the -5% scenario for conservative procurement, inventory and staffing planning.

4. Track actual-versus-forecast performance weekly and retrain the model when accuracy deteriorates.

5. Improve the next model version by adding external drivers such as weather, holidays, outages, pricing and other demand drivers when available.

5. Limitations and Assumptions

This model is based only on historical demand. It does not explicitly model weather, holidays, outages, price changes or other exogenous events. Therefore, the forecast should be treated as a planning baseline rather than a guaranteed outcome. The ±5% scenarios are management planning assumptions, not statistically estimated scenarios.

Submission package includes the reproducible notebook, cleaned datasets, forecast CSV and supporting charts.